



Department of Computer Science & Engineering

Course Code: CSE 4346

Course Name: Big Data Analytics Laboratory

Project Report

On

“Big Data Analytics of E-Commerce Sales Transactions Using
Hadoop MapReduce and Apache Pig”

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Batch: 42nd

Semester: 8th , Section: B

Session: Spring 2026

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Submission Date: 13th September, 2026

Big Data Analytics of E-Commerce Sales Transactions Using Hadoop MapReduce and Apache Pig

Abstract

E-commerce platforms generate continuous streams of transactional data covering orders, customers, products, payments, and delivery, and this volume only grows as a business scales. Unlike traditional retail, where a transaction happens face to face and much of the surrounding context is never recorded, an e-commerce order captures dozens of attributes automatically from device type and traffic source to delivery status and customer rating which makes the resulting datasets both larger and richer than their offline counterparts. This project applies the core Hadoop ecosystem the Hadoop Distributed File System (HDFS), MapReduce, and Apache Pig to a dataset of over one million e-commerce order records. HDFS is used to store the dataset in a distributed and fault-tolerant manner, splitting it into blocks across the cluster rather than relying on a single disk. Four MapReduce jobs, written in Java, compute foundational aggregates: total order count, category-wise sales, average order value by country, and total profit by payment method. Ten Apache Pig scripts then extend this analysis to cover payment distribution, monthly sales trends, best-selling products, top revenue countries, brand profitability, and delivery performance. Two of these analyses average order value by country were implemented independently in both MapReduce and Pig, allowing the results of one to be cross-checked against the other. The project demonstrates, end to end, how distributed storage and parallel processing can be used to turn a large raw transactional dataset into structured business insight, and compares the practical trade-offs between writing analysis logic by hand in Java MapReduce versus expressing the same logic in Pig Latin.

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1. Introduction

E-commerce platforms record every stage of a transaction — browsing, ordering, payment, and delivery — resulting in datasets that are too large and too varied for a single machine or a conventional relational database to process efficiently. As order volumes climb into the hundreds of thousands or millions, queries that would return instantly on a small spreadsheet start to take minutes or hours on a single machine, and the risk of losing data to a single point of failure becomes a genuine operational concern rather than a theoretical one. This project works with a dataset of order-level e-commerce records containing customer demographics, product details, pricing, payment method, shipping information, and delivery outcomes, giving a realistic picture of what a mid-sized online retailer's raw transaction log actually looks like in practice.

The objective is to apply distributed Big Data tools to this dataset in three complementary ways. First, HDFS is used to store the raw dataset reliably across the (pseudo-distributed) Hadoop cluster, so that the data survives node failure and can be read in parallel by multiple tasks at once. Second, MapReduce is used to implement four core aggregation jobs directly in Java, giving fine-grained control over exactly how each record is read, mapped, and reduced. Third, Apache Pig is used to perform ten further analyses using Pig Latin, which expresses the same style of group-by/aggregate logic far more concisely than hand-written MapReduce code, while still compiling down to MapReduce jobs behind the scenes.

2. Problem Statement

A single e-commerce order record is simple, but a merchant's real question is rarely about one order it is about patterns across a million of them: which categories sell best, which countries generate the most revenue, which payment methods are most profitable, and where delivery is under-performing. These are exactly the kinds of questions that decide inventory planning, marketing spend, and logistics partnerships in a real business, so answering them accurately matters well beyond the technical exercise itself. Answering these questions over a dataset of this size with a conventional single-machine script becomes slow and memory-constrained as the data grows, and the problem only worsens as more months of orders are added to the pile.

This project addresses that problem by distributing both storage and computation. The dataset is stored in HDFS so that it is split into blocks across the cluster rather than held on one disk, which also protects against data loss through replication across nodes. MapReduce and Apache Pig are then used to run the actual aggregation logic in parallel across those blocks, so that operations such as counting orders, summing revenue by category, or averaging order value by country scale with the cluster rather than with a single CPU. In this sense, the problem being solved is not just producing the right numbers once, but producing them in a way that keeps working as the dataset keeps growing.

3. Dataset Description

The dataset used is `ecommerce_dataset_+1m.csv`, containing just over one million e-commerce order records collected from Kaggle. Each row describes a single order and includes fields such as:

- Order details — order_id, order_date, order_year/month/day/hour/minute/second, is_weekend, order_status, return_reason
- Customer details — customer_id, customer_name, gender, age, customer_segment, country, city, customer_loyalty_score, total_orders_by_customer, account_creation_date
- Product details — product_id, product_name, category, sub_category, brand, product_rating_avg, product_reviews_count, stock_quantity
- Order economics — unit_price_usd, quantity, discount_percent, discount_amount_usd, total_price_usd, cost_usd, profit_usd, tax_usd, currency
- Payment and shipping — payment_method, payment_status, installment_plan, shipping_method, shipping_cost_usd, delivery_days, shipping_country, warehouse_location, delivery_status
- Customer experience — rating, review_sentiment, customer_feedback, coupon_used, coupon_code, campaign_source, device_type, traffic_source, session_duration_minutes, pages_visited, abandoned_cart_before, fraud_risk_score, profit_margin_percent, order_priority, support_ticket_created

Dataset: <https://www.kaggle.com/datasets/akrambelha/global-e-commerce-dataset-1m-records-20242026>

4. Setting Up the Working Environment

Before any HDFS, MapReduce, or Pig work could begin, the required tools had to be installed and verified from the command line, since a missing or misconfigured tool at this stage would cause confusing failures much later in the pipeline:

- Java (JDK) — required since Hadoop and MapReduce programs run on the JVM.
- Hadoop 3.2.4 — verified with `hadoop version`, with `java -version` and `hadoop version` set so Hadoop commands run from any Command Prompt.
- Apache Pig 0.18.0 — verified with `pig -version`.

If all these commands return valid version information without errors, the environment is confirmed ready, and the project can proceed to starting the Hadoop services and preparing the dataset.

```
C:\Windows\system32>java -version
java version "1.8.0_411"
Java(TM) SE Runtime Environment (build 1.8.0_411-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.411-b09, mixed mode)

C:\Windows\system32>hadoop version
Hadoop 3.2.4
Source code repository Unknown -r 7e5d9983b388e372fe640f21f048f2f2ae6e9eba
Compiled by ubuntu on 2022-07-12T11:58Z
Compiled with protoc 2.5.0
From source with checksum ee031c16fe785bbb35252c749418712
This command was run using /C:/hadoop/share/hadoop/common/hadoop-common-3.2.4.jar

C:\Windows\system32>pig -version
Apache Pig version 0.18.0 (r1926484)
compiled Jun 16 2025, 10:38:21

C:\Windows\system32>
```

5. Storing the E-Commerce Data in HDFS

5.1 Launching the Hadoop Services

This step starts the Hadoop services so that HDFS and YARN are running and ready to store and process data. HDFS and YARN were started so that the cluster could store and process data. Running start-dfs.cmd and start-yarn.cmd launches the required daemons, and jps confirms that NameNode, DataNode, ResourceManager, and NodeManager are all running before any further commands are attempted.

```
C:\hadoop\sbin>start-dfs.cmd  
  
C:\hadoop\sbin>start-yarn.cmd  
starting yarn daemons  
  
C:\hadoop\sbin>jps  
10132 NodeManager  
10696 DataNode  
280 Jps  
9320 NameNode  
4828 ResourceManager
```

5.2 Create a Project Directory

A local project folder, C:\ECommerceBigData, was created with separate subfolders to keep the dataset, MapReduce source code, compiled classes, packaged JAR, and Pig scripts organised and isolated from one another.

```
C:\hadoop\sbin>cd C:\  
  
C:\>mkdir ECommerceBigData  
  
C:\>cd ECommerceBigData  
  
C:\ECommerceBigData>mkdir dataset  
  
C:\ECommerceBigData>mkdir java  
  
C:\ECommerceBigData>mkdir classes  
  
C:\ECommerceBigData>mkdir jar  
  
C:\ECommerceBigData>mkdir pig
```

5. Dataset Upload and Check

To make the dataset available for distributed processing in Hadoop, the local file was uploaded to the Hadoop Distributed File System (HDFS)

File Presence Check:

```
C:\ECommerceBigData>dir C:\ECommerceBigData\dataset  
Volume in drive C is SYSTEM  
Volume Serial Number is 8AEA-60C9
```

Integrity & Size Verification: The file size was checked (~410 MB) to confirm the dataset was complete and had not been corrupted or truncated before initiating the HDFS transfer.

```
C:\ECommerceBigData>dir C:\ECommerceBigData\dataset
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\dataset

09/10/2026  08:56 PM    <DIR>          .
09/10/2026  08:56 PM    <DIR>          ..
02/03/2026  11:28 AM             410,450,731 ecommerce_dataset_+1m.csv
               1 File(s)      410,450,731 bytes
               2 Dir(s)      80,965,586,944 bytes free
```

5.4 Building the HDFS Directory Structure

The commands create separate HDFS folders for raw input and generated MapReduce/Pig results, matching the local project structure.

```
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/input
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output/mapreduce
C:\ECommerceBigData>hdfs dfs -mkdir /ecommerce/output/pig
```

5.5 Copying the Dataset into HDFS

This step transfers the dataset from the local file system into HDFS so that it can be accessed and processed in a distributed manner by MapReduce and Pig.

```
C:\ECommerceBigData>hdfs dfs -put "C:\ECommerceBigData\dataset\ecommerce_dataset_+1m.csv" /ecommerce/input/
C:\ECommerceBigData>hdfs dfs -ls /ecommerce/input
Found 1 items
-rw-r--r--  1 Anamika supergroup  410450731 2026-09-10 21:01 /ecommerce/input/ecommerce_dataset_+1m.csv
```

5.6 Verifying the Uploaded Dataset

File Size Verification:

```
C:\ECommerceBigData\pig>hdfs dfs -du -h /ecommerce/input/ecommerce_dataset_+1m.csv
391.4 M  391.4 M  /ecommerce/input/ecommerce_dataset_+1m.csv
```

Record Count Validation: The command returned a line count of 1,000,124, confirming that the dataset contains 1,000,123 transactional records alongside the single CSV header row.

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/input/ecommerce_dataset_+1m.csv | find /c /v ""
1000124
```

Schema & Header Inspection: The `hdfs dfs -cat` command with PowerShell read the first line, confirming the header and all 62 columns were loaded correctly into HDFS.

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/input/ecommerce_dataset_+1m.csv | powershell -Command "$input | Select-Object -First 1"
order_id,order_date,order_year,order_month,order_day,order_hour,order_minute,order_second,is_weekend,order_status,return_reason,customer_id,customer_name,gender,age,customer_segment,country,city,customer_loyalty_score,total_orders_by_customer,account_creation_date,product_id,product_name,category,sub_category,brand,product_rating_avg,product_reviews_count,stock_quantity,unit_price_usd,quantity,discount_percent,discount_amount_usd,total_price_usd,cost_usd,profit_usd,tax_usd,currency,payment_method,payment_status,installment_plan,shipping_method,shipping_cost_usd,delivery_days,shipping_country,warehouse_location,delivery_status,rating,review_sentiment,customer_feedback,coupon_used,coupon_code,campaign_source,device_type,traffic_source,session_duration_minutes,pages_visited,abandoned_cart_before,fraud_risk_score,profit_margin_percent,order_priority,support_ticket_created
cat: Unable to write to output stream.
```

6. Distributed Processing Using MapReduce

Four MapReduce jobs were implemented in Java, each with its own Driver, Mapper, and Reducer class, and packaged together into a single JAR (ecommerce.jar). Building all four into one JAR meant that only one file needed to be submitted to Hadoop for every job, rather than rebuilding a separate package each time a new analysis was added.

6.1 Writing the Driver, Mapper and Reducer Classes

Each job uses three Java classes: a Mapper, Reducer, and Driver, following the standard MapReduce structure.

```
C:\ECommerceBigData>dir java
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\java

09/10/2026  09:07 PM    <DIR>          .
09/10/2026  09:07 PM    <DIR>          ..
09/10/2026  09:07 PM                965 Driver1.java
09/10/2026  09:06 PM                855 Mapper1.java
09/10/2026  09:06 PM                556 Reducer1.java
               3 File(s)                2,376 bytes
               2 Dir(s)  80,546,480,128 bytes free
```

6.2 Compile the MapReduce Classes Folder

The three Java files are compiled using javac (with the Hadoop classpath included). Compiling here catches any syntax or dependency errors before the job is packaged and run.

```
C:\ECommerceBigData>dir classes
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\classes

09/10/2026  09:10 PM    <DIR>          .
09/10/2026  09:10 PM    <DIR>          ..
09/10/2026  09:10 PM                1,349 Driver1.class
09/10/2026  09:10 PM                1,862 Mapper1.class
09/10/2026  09:10 PM                1,586 Reducer1.class
               3 File(s)                4,797 bytes
               2 Dir(s)  80,548,618,240 bytes free
```

6.3 Compiling the Java Source

Each class was compiled using **javac** with the Hadoop Common and MapReduce libraries in the classpath. This ensured Hadoop-specific types resolved correctly and helped catch syntax or type errors before job submission.

```
C:\ECommerceBigData>hadoop classpath
C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\common;C:\hadoop\share\hadoop\common\lib\*;C:\hadoop\share\hadoop\common\*;C:\hadoop\share\hadoop\hdfs;C:\hadoop\share\hadoop\hdfs\lib\*;C:\hadoop\share\hadoop\hdfs\*;C:\hadoop\share\hadoop\yarn;C:\hadoop\share\hadoop\yarn\lib\*;C:\hadoop\share\hadoop\yarn\*;C:\hadoop\share\hadoop\mapreduce\lib\*;C:\hadoop\share\hadoop\mapreduce\*

C:\ECommerceBigData>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\common\lib\*;%HADOOP_HOME%\share\hadoop\mapreduce\*;%HADOOP_HOME%\share\hadoop\mapreduce\lib\*" -d classes java\Mapper1.java java\Reducer1.java java\Driver1.java
```

6.4 Building the JAR File

All twelve .class files were packaged into a single ecommerce.jar using the `jar` command. This JAR file was then submitted to Hadoop using `hadoop jar`.

```
C:\ECommerceBigData>cd classes

C:\ECommerceBigData\classes>jar -cvf ecommerce.jar *.class
added manifest
adding: Driver1.class(in = 1349) (out= 751)(deflated 44%)
adding: Mapper1.class(in = 1862) (out= 788)(deflated 57%)
adding: Reducer1.class(in = 1586) (out= 662)(deflated 58%)
```

```
C:\ECommerceBigData\classes>cd ..

C:\ECommerceBigData>dir classes\ecommerce.jar
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\classes

09/10/2026  09:23 PM                2,899 ecommerce.jar
               1 File(s)                2,899 bytes
               0 Dir(s)  80,320,385,024 bytes free
```

6.5 Run the MapReduce

Four MapReduce jobs were implemented they are:

- **Job 1-OrderCount** = Total Order Count
- **Job 2-CategorySales** = Category-wise Total Sales
- **Job 3-AvgOrderValue** = Average Order Value by Country
- **Job 4-ProfitByPaymentMethod** = Total Profit by Payment Method

Browse Directory

/ecommerce/output/mapreduce

Go!



Show

25

 entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	☐
☐	drwxr-xr-x	Anamika	supergroup	0 B	Sep 10 21:28	0	0 B	Job1-TotalOrderCount	
☐	drwxr-xr-x	Anamika	supergroup	0 B	Sep 10 21:48	0	0 B	Job2-CategoryWiseSales	
☐	drwxr-xr-x	Anamika	supergroup	0 B	Sep 10 21:51	0	0 B	Job3-AvgOrderValueByCountry	
☐	drwxr-xr-x	Anamika	supergroup	0 B	Sep 10 21:54	0	0 B	Job4-TotalProfitByPaymentMethod	

Showing 1 to 4 of 4 entries

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6.5.1 Job 1-Counting Total Orders

This job counts the total number of valid order records in the dataset. The Mapper emits a constant key with a value of 1 for each order; the Reducer sums these to a single total. Although conceptually the simplest of the four jobs, it establishes a baseline figure that every later per-category or per-country percentage can be compared against.

6.5.1.1 Run the MapReduce job

The job is submitted using `hadoop jar`, pointing to the input dataset in HDFS and specifying an output directory for the results. The Mapper reads each record and emits a constant key with a

count of 1 per valid order record, while the Reducer sums these counts to produce the final overall total.

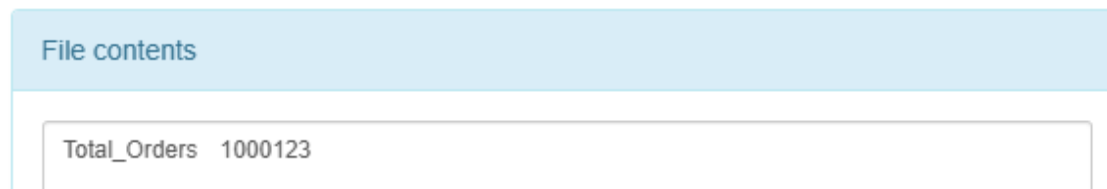
```
C:\ECommerceBigData>hadoop jar classes\ecommerce.jar Driver1 /ecommerce/input/ecommerce_dataset_1m.csv /ecommerce/output/mapreduce/job1
2026-09-10 21:28:01,230 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:28:02,044 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:28:02,030 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamikaa/.staging/job_1789050973944_0001
2026-09-10 21:28:02,225 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 21:28:02,301 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:28:02,414 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0001
2026-09-10 21:28:02,416 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:28:02,579 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:28:02,580 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 21:28:02,991 INFO impl.YarnClientImpl: Submitted application application_1789050973944_0001
2026-09-10 21:28:03,032 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8088/proxy/application_1789050973944_0001/
2026-09-10 21:28:03,033 INFO mapreduce.Job: Running job: job_1789050973944_0001
2026-09-10 21:28:16,234 INFO mapreduce.Job: Job job_1789050973944_0001 running in uber mode : false
2026-09-10 21:28:16,237 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:28:26,408 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:28:30,481 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:28:35,601 INFO mapreduce.Job: Job job_1789050973944_0001 completed successfully
2026-09-10 21:28:35,685 INFO mapreduce.Job: Counters: 54
  File System Counters
    FILE: Number of bytes read=19002343
    FILE: Number of bytes written=38959507
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=410459307
    HDFS: Number of bytes written=21
    HDFS: Number of read operations=14
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=3
    Launched reduce tasks=1
    Data-local map tasks=3
    Total time spent by all maps in occupied slots (ms)=17238
    Total time spent by all reduces in occupied slots (ms)=4273
    Total time spent by all map tasks (ms)=17238
```

6.5.1.2 Check the Result Directory

```
C:\ECommerceBigData>hdfs dfs -ls /ecommerce/output/mapreduce/job1
Found 2 items
-rw-r--r-- 1 Anamikaa supergroup          0 2026-09-10 21:28 /ecommerce/output/mapreduce/job1/_SUCCESS
-rw-r--r-- 1 Anamikaa supergroup       21 2026-09-10 21:28 /ecommerce/output/mapreduce/job1/part-r-00000
```

```
C:\ECommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job1/part-r-00000
Total_Orders  1000123
```

6.5.1.3 Display the Final Output



6.5.2 Job 2-Sales Totals per Category

This job groups orders by product category and sums total sales, identifying the categories that generate the most revenue.

6.5.2.1 Run the MapReduce Job

```
C:\ECommerceBigData>hadoop jar classes\ecommerce.jar Driver2 /ecommerce/input/ecommerce_dataset_1m.csv /ecommerce/output/mapreduce/job2
2026-09-10 21:47:50,799 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:47:51,549 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:47:51,560 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamikaa/.staging/job_1789050973944_0002
2026-09-10 21:47:51,752 INFO input.FileInputFormat: total input files to process : 1
2026-09-10 21:47:51,910 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:47:51,904 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0002
2026-09-10 21:47:52,088 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:47:52,088 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:47:52,088 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 21:47:52,441 INFO impl.YarnClientImpl: Submitted application application_1789050973944_0002
2026-09-10 21:47:52,174 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8088/proxy/application_1789050973944_0002/
2026-09-10 21:47:52,174 INFO mapreduce.Job: Running job: job_1789050973944_0002
2026-09-10 21:48:04,380 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:48:04,380 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:48:19,640 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:48:24,726 INFO mapreduce.Job: Job job_1789050973944_0002 completed successfully
2026-09-10 21:48:24,684 INFO mapreduce.Job: Counters: 55
  File System Counters
    FILE: Number of bytes read=18000102
    FILE: Number of bytes written=38959509
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=410459307
    HDFS: Number of bytes written=142
    HDFS: Number of read operations=14
    HDFS: Number of large read operations=0
```

6.5.2.2 Check the Result Directory

```
C:\ECommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job2/part-r-00000
Clothing      4.5205795899999134E7
Electronics   1.3986598693999827E8
Health        5.746792554999967E7
Home          8.782867776000017E7
Sports        7.291747085000059E7
```

6.5.2.3 Display the Final Output

File contents	
Clothing	4.5205795899999134E7
Electronics	1.3986598693999827E8
Health	5.746792554999967E7
Home	8.782867776000017E7
Sports	7.291747085000059E7

6.5.3 Job 3-Average Order Value per Country

This job groups orders by country and calculates average spending, requiring both total sum and order count to produce an accurate distributed average.

6.5.3.1 Run the MapReduce Job

```
C:\ECommerceBigData>hadoop jar classes\ecommerce.jar Driver3 /ecommerce/input/ecommerce_dataset_1m.csv /ecommerce/output/mapreduce/job3
2026-09-10 21:51:05.128 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:51:05.910 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:51:06.924 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamika/.staging/job_1789050973944_0003
2026-09-10 21:51:06.109 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 21:51:06.168 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:51:06.266 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0003
2026-09-10 21:51:06.268 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:51:06.429 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:51:06.429 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'
2026-09-10 21:51:06.482 INFO impl.YarnClientImpl: Submitted application application_1789050973944_0003
2026-09-10 21:51:06.515 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8088/proxy/application_1789050973944_0003/
2026-09-10 21:51:06.516 INFO mapreduce.Job: Running job: job_1789050973944_0003
2026-09-10 21:51:18.736 INFO mapreduce.Job: Job job_1789050973944_0003 running in uber mode : false
2026-09-10 21:51:18.737 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:51:26.081 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:51:33.963 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:51:39.045 INFO mapreduce.Job: Job job_1789050973944_0003 completed successfully
2026-09-10 21:51:39.127 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=19303086
  FILE: Number of bytes written=39561057
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=410459307
  HDFS: Number of bytes written=278
  HDFS: Number of read operations=14
```

6.5.3.2 Check the Result Directory

```
C:\ECommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job3/part-r-00000
Australia      404.2269420891208
Belgium        402.7290564684118
Canada         401.9354866380752
France         401.71363695367967
Germany        402.63543595609747
Italy          405.7300347831337
Netherlands    403.24070583203815
Spain          403.6074160594136
United Kingdom 403.14696014550543
United States  403.40777806741994
```

6.5.3.3 Display the Final Output

File contents	
Australia	404.2269420891208
Belgium	402.7290564684118
Canada	401.9354866380752
France	401.71363695367967
Germany	402.63543595609747
Italy	405.7300347831337
Netherlands	403.24070583203815
Spain	403.6074160594136

6.5.4 Job 4-Profit Totals per Payment Method

Groups orders by payment_method and sums profit_usd to identify which payment method contributes the most to overall profitability.

6.5.4.1 Run the MapReduce Job

```
C:\EcommerceBigData>hadoop jar classes\ecommerce.jar Driver4 /ecommerce/input/ecommerce_dataset_+im.csv /ecommerce/output/mapreduce/job4
2026-09-10 21:54:14,188 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 21:54:14,934 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application
with ToolRunner to remedy this.
2026-09-10 21:54:14,948 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anamika/.staging/job_1789050973944_0004
2026-09-10 21:54:15,129 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 21:54:15,187 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 21:54:15,283 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789050973944_0004
2026-09-10 21:54:15,285 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 21:54:15,445 INFO conf.Configuration: resource-types.xml not found
2026-09-10 21:54:15,446 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 21:54:15,500 INFO impl.YarnClientImpl: Submitted application application_1789050973944_0004
2026-09-10 21:54:15,533 INFO mapreduce.Job: The url to track the job: http://DESKTOP-OKPREUF:8088/proxy/application_1789050973944_0004/
2026-09-10 21:54:15,534 INFO mapreduce.Job: Running job: job_1789050973944_0004
2026-09-10 21:54:27,737 INFO mapreduce.Job: Job job_1789050973944_0004 running in uber mode : false
2026-09-10 21:54:27,740 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 21:54:35,886 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 21:54:42,963 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 21:54:48,055 INFO mapreduce.Job: Job job_1789050973944_0004 completed successfully
2026-09-10 21:54:48,138 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=20803300
  FILE: Number of bytes written=42561485
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=410459307
  HDFS: Number of bytes written=159
  HDFS: Number of read operations=14
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=3
  Launched reduce tasks=1
  Data-Local map tasks=3
```

6.5.4.2 Check the Result Directory

```
C:\EcommerceBigData>hdfs dfs -cat /ecommerce/output/mapreduce/job4/part-r-00000
Apple Pay      3.2072775629999757E7
Bank Transfer  3.2250553240000267E7
Credit Card   3.2119889630000137E7
Debit Card     3.2356712459999777E7
PayPal         3.2146881920000304E7
```

6.5.4.3 Display the Final Output

File contents

```
Apple Pay      3.2072775629999757E7
Bank Transfer  3.2250553240000267E7
Credit Card   3.2119889630000137E7
Debit Card     3.2356712459999777E7
PayPal         3.2146881920000304E7
```

6.5.5 Managing Java Source Files and Directory Structure

Directory listing of the java folder showing Java source files for all four MapReduce jobs.

```

C:\ECommerceBigData>dir java
Volume in drive C is SYSTEM
Volume Serial Number is 8AEA-60C9

Directory of C:\ECommerceBigData\java

09/10/2026  09:43 PM    <DIR>          .
09/10/2026  09:43 PM    <DIR>          ..
09/10/2026  09:07 PM             965 Driver1.java
09/10/2026  09:38 PM             921 Driver2.java
09/10/2026  09:42 PM             926 Driver3.java
09/10/2026  09:43 PM             926 Driver4.java
09/10/2026  09:06 PM             855 Mapper1.java
09/10/2026  09:36 PM             834 Mapper2.java
09/10/2026  09:40 PM             817 Mapper3.java
09/10/2026  09:42 PM             821 Mapper4.java
09/10/2026  09:06 PM             556 Reducer1.java
09/10/2026  09:37 PM             494 Reducer2.java
09/10/2026  09:41 PM             595 Reducer3.java
09/10/2026  09:43 PM             494 Reducer4.java
               12 File(s)          9,204 bytes
               2 Dir(s)      80,406,986,752 bytes free

```

Compiling the Java Source

```

C:\ECommerceBigData>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\common\lib\*;%HADOOP_HOME%\share\hadoop\mapreduce\*;%HADOOP_HOME%\share\hadoop\mapreduce\lib\*" -d classes java\Mapper2.java java\Reducer2.java java\Driver2.java java\Mapper3.java java\Reducer3.java java\Driver3.java java\Mapper4.java java\Reducer4.java java\Driver4.java

```

Building the JAR File

```

C:\ECommerceBigData>cd classes
C:\ECommerceBigData\classes>jar -cvf ecommerce.jar *.class
added manifest
adding: Driver1.class(in = 1349) (out= 751)(deflated 44%)
adding: Driver2.class(in = 1360) (out= 760)(deflated 44%)
adding: Driver3.class(in = 1365) (out= 763)(deflated 44%)
adding: Driver4.class(in = 1365) (out= 764)(deflated 44%)
adding: Mapper1.class(in = 1862) (out= 788)(deflated 57%)
adding: Mapper2.class(in = 1903) (out= 796)(deflated 58%)
adding: Mapper3.class(in = 1903) (out= 796)(deflated 58%)
adding: Mapper4.class(in = 1903) (out= 796)(deflated 58%)
adding: Reducer1.class(in = 1586) (out= 662)(deflated 58%)
adding: Reducer2.class(in = 1604) (out= 663)(deflated 58%)
adding: Reducer3.class(in = 1643) (out= 697)(deflated 57%)
adding: Reducer4.class(in = 1604) (out= 664)(deflated 58%)

```

7. Extended Analysis Using Apache Pig

Apache Pig was used to extend the analysis beyond the four MapReduce jobs, using Pig Latin scripts that load, clean, group, and aggregate the same dataset with considerably less code than the equivalent Java would require. All scripts were stored under the local pig folder and run with `pig -x mapreduce <script>.pig`, with results stored in HDFS under `/ecommerce/output/pig`. Because Pig compiles each script down into one or more MapReduce jobs automatically, the underlying execution model is identical only the amount of code the analyst has to write by hand changes.

7.1 Create the Pig Script Folder

All ten scripts sit directly inside the pig folder, distinguished by descriptive, numbered file names:

- `pig1_rowcheck.pig` — data loading and cleaning (row count check)
- `pig2_ordercount.pig` — total order count
- `pig3_categorysales.pig` — category-wise sales
- `pig4_countryavg.pig` — average order value by country
- `pig5_paymentdist.pig` — payment method distribution
- `pig6_monthlytrend.pig` — monthly sales trend
- `pig7_topproducts.pig` — top 10 best-selling products
- `pig8_topcountries.pig` — top 10 countries by revenue

- pig9_brandprofit.pig — brand-wise total profit
- pig10_deliverystatus.pig — delivery status breakdown

Browse Directory

/ecommerce/output/pig

Go!

Show

25

entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:13	0	0 B	AvgOrderValueByCountry	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 15:13	0	0 B	BrandWiseTotalProfit	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 13:59	0	0 B	CategoryWiseSales	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 15:17	0	0 B	DeliveryStatusBreakdown	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:19	0	0 B	PaymentMethodDistribution	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 13:43	0	0 B	RowCountCheck	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:25	0	0 B	RowCountCheckMonthlySalesTrend	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 14:57	0	0 B	Top10BestSellingProducts	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 15:07	0	0 B	Top10CountriesByRevenue	
☐	drwxr-xr-x	Anamikaa	supergroup	0 B	Sep 11 13:52	0	0 B	TotalOrderCount	

Showing 1 to 10 of 10 entries

Previous

1

Next

7.2 Pig analysis

7.2.1 Pig Data Loading & Cleaning

Every script loads the CSV with a defined schema, removes the header, and filters records missing order_id or total_price_usd. The pig1_rowcheck.pig script compares raw and cleaned row counts to verify data quality.

7.2.1.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData\pig> pig -x mapreduce pig1_rowcheck.pig
2026-09-11 13:12:49,628 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 13:12:49,631 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 13:12:49,631 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 13:12:50,317 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 13:12:50,318 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789110770306.log
2026-09-11 13:12:51,209 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\USER\pigbootstrap not found
2026-09-11 13:12:51,397 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 13:12:51,397 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 13:12:52,508 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig1_rowcheck.pig-b19ecd1-4c2d-4586-89c2-f9996f07ffe0
2026-09-11 13:12:52,508 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 13:12:54,091 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP BY
2026-09-11 13:12:54,129 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 13:12:54,184 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForeachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCasterInserter, MergeFilter, MergeForEach, NestedInitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCasterInserter]]
2026-09-11 13:12:54,325 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489508128, usageThreshold = 489508128
2026-09-11 13:12:54,467 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 13:12:54,501 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 13:12:54,558 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 13:12:54,558 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 13:12:54,738 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 13:12:55,270 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 13:12:55,298 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 13:12:55,305 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 13:12:55,312 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 13:12:55,312 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 13:12:55,319 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 13:12:55,321 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-11 13:12:55,325 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 13:12:55,326 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 13:12:55,361 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 13:12:55,647 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduceLayer.JobControlCompiler - Added jar file: C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/tmp572521480/tmp282923997/pig-0.18.0-core-h3.jar
```

7.2.1.2 Show the actual output

```
C:\ECommerceBigData\pig> hdfs dfs -cat /ecommerce/output/pig/analysis1/part-m-00000
Cleaned Row Count,1000123

C:\ECommerceBigData\pig> hdfs dfs -cat /ecommerce/output/pig/analysis1/part-m-00001
Raw Row Count,1000124

C:\ECommerceBigData\pig> hdfs dfs -cat /ecommerce/output/pig/analysis1/part-*
Cleaned Row Count,1000123
Raw Row Count,1000124
```

7.2.2 Counting Total Orders in Pig

Groups all cleaned records and counts them, confirming the same data processing as MapReduce Job 1.

7.2.2.1 Create the Pig script

```
C:\ECommerceBigData\pig>notepad pig2_ordercount.pig
```

7.2.2.2 Run the Pig script

[illegible]

7.2.2.3 Show the actual output

```
C:\ECommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/analysis2/part-r-000001000123
```

7.2.3 Sales Totals per Category in Pig

Groups records by category and calculates order count and total sales together, helping compare whether high revenue comes from many small orders or fewer high-value orders.

7.2.3.1 Run the Pig script

[illegible]

7.2.3.2 Show the actual output

File contents

```
Home,200589,8.782867776000027E7
Health,199574,5.746792554999983E7
Sports,200397,7.291747085000002E7
Clothing,199544,4.5205795900000416E7
Electronics,200019,1.398659869399997E8
```

7.2.4 Average Order Value by Country in Pig

Groups records by country and calculates the average order value using AVG, providing the same result.

7.2.4.1 Create the Pig file and Run the Pig script

```
C:\EcommerceBigData\pig>notepad pig4_countryavg.pig
C:\EcommerceBigData\pig>pig -x mapreduce pig4_countryavg.pig
2026-09-11 14:12:31,003 INFO org.apache.pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:12:31,070 INFO org.apache.pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:12:31,071 INFO org.apache.pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:12:31,966 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:12:31,966 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig 1789114351953.log
2026-09-11 14:12:33,182 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\AUSER\pigbootstrap not found
2026-09-11 14:12:33,251 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:12:33,251 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:12:33,957 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig4_countryavg.pig-ede23500-a853-41ed-9113-bc06e24e6783
2026-09-11 14:12:33,957 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:12:35,159 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.t
extoutputformat.separator
2026-09-11 14:12:35,171 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 14:12:35,203 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 14:12:35,259 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[addfortach, ColumnMapKeyPrune, FilterConstantCalcul
ator, ForceConstantCalculator, GroupConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergerForEach, NestedLimitOptimizer, PartitionFilterOptim
izer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:12:35,378 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 6994808192 to monitor. collectionUsageThreshold
= 489588128, usageThreshold = 489588128
2026-09-11 14:12:35,500 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:12:35,528 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:12:35,550 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 14:12:35,560 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 14:12:35,600 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:12:35,903 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, us
e yarn.system-metrics-publisher.enabled
2026-09-11 14:12:35,970 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:12:35,978 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce
.reduce.markreset.buffer.percent
2026-09-11 14:12:35,979 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
set, set to default 0.3
2026-09-11 14:12:35,986 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputfo
rmat.compress
2026-09-11 14:12:35,990 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required r
educers.
2026-09-11 14:12:35,992 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.h
adoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-11 14:12:36,003 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999
totalInputFileSize=410450731
```

7.2.4.2 Show the actual output

```
C:\EcommerceBigData\pig>hdfs dfs -cat /ecommerce/output/pig/AvgOrderValueByCountry/part-r-00000
Italy,405.7300347831321
Spain,403.60741605941524
Canada,401.9354866380731
France,401.7136369536785
Belgium,402.72905646840763
Germany,402.63543595609656
Australia,404.22694208912384
Netherlands,403.2407058320406
United States,403.4077780674112
United Kingdom,403.14696014550645
```

7.2.5 Payment Method Distribution in Pig

Groups records by payment method and calculates order count and total revenue, helping compare popular payment methods with their revenue performance.

7.2.5.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData>pignotepad pig_paymentdist.pig
C:\ECommerceBigData>pig -x mapreduce pig_paymentdist.pig
2026-09-11 14:18:42,611 [main] INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:18:42,614 [main] INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:18:42,615 [main] INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:18:43,460 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1026484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:18:43,470 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1789114723455-log
2026-09-11 14:18:44,671 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 14:18:44,752 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:18:44,752 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:18:45,479 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: Pig-pig_paymentdist.pig-6509f65-65ce-4b02-b3ee-78feda5c674
2026-09-11 14:18:45,479 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:18:46,612 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.t
extoutputformat.separator
2026-09-11 14:18:46,634 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP BY,FILTER
2026-09-11 14:18:46,662 [main] INFO org.apache.pig.data.SchemaTupleBackend - key [pig.schematuple] was not set... Will not generate code.
2026-09-11 14:18:46,701 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED:[addForEach, ColumnMapKeyPrune, FilterConstantCalcul
ator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptim
izer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:18:46,776 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold
= 489580128, usageThreshold = 489580128
2026-09-11 14:18:46,879 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:18:46,920 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:18:46,980 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 14:18:46,981 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 14:18:47,138 [main] INFO org.apache.hadoop.yarn.client.RetryProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:18:47,398 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, us
e yarn.system-metrics-publisher.enabled
2026-09-11 14:18:47,414 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:18:47,421 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce
.reduce.markreset.buffer.percent
2026-09-11 14:18:47,422 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
set, set to default 0.3
2026-09-11 14:18:47,423 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputfo
rmat.compress
2026-09-11 14:18:47,429 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required r
educers.
```

7.2.5.2 Show the actual output

File contents

```
PayPal,199881.8,05951279600001E7
Apple Pay,199814.8,039556929000005E7
Debit Card,200805.8,106516731000024E7
Credit Card,199410.8,041428974000017E7
Bank Transfer,200213.8,081570269999954E7
```

7.2.6 Monthly Sales Trend in Pig

Groups records by month, calculates order count and total sales, and sorts them chronologically to identify seasonal demand patterns.

7.2.6.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData>pignotepad pig_monthlytrend.pig
C:\ECommerceBigData>pig -x mapreduce pig_monthlytrend.pig
2026-09-11 14:22:11,528 [main] INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:22:11,521 [main] INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:22:11,523 [main] INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:22:12,163 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:22:12,164 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1789114932148-log
2026-09-11 14:22:13,245 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 14:22:13,314 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:22:13,314 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:22:14,138 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: Pig-pig_monthlytrend.pig-0b22e3d5-e7de-4a73-b080-0534ead7ac5
2026-09-11 14:22:14,138 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:22:15,321 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.t
extoutputformat.separator
2026-09-11 14:22:15,343 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP BY,ORDER BY,FILTER
2026-09-11 14:22:15,373 [main] INFO org.apache.pig.data.SchemaTupleBackend - key [pig.schematuple] was not set... Will not generate code.
2026-09-11 14:22:15,429 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED:[addForEach, ColumnMapKeyPrune, FilterConstantCalcul
ator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptim
izer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:22:15,544 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold
= 489580128, usageThreshold = 489580128
2026-09-11 14:22:15,679 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:22:15,764 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:22:15,796 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapRedu
ce node scope-228
2026-09-11 14:22:15,816 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-11 14:22:15,816 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-11 14:22:15,956 [main] INFO org.apache.hadoop.yarn.client.RetryProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:22:16,251 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, us
e yarn.system-metrics-publisher.enabled
2026-09-11 14:22:16,267 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:22:16,276 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce
.reduce.markreset.buffer.percent
2026-09-11 14:22:16,277 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not
set, set to default 0.3
2026-09-11 14:22:16,279 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputfo
rmat.compress
```

7.2.6.2 Show the actual output

```
C:\ECommerceBigData>hdfs dfs -cat /eCommerce/output/pig/MonthlySalesTrend2/part-r-00000
1,85100,3,4410154269999995E7
2,77685,3,139802869000005E7
3,85079,3,418629474000016E7
4,82436,3,3343639369999915E7
5,85100,3,4282929650000095E7
6,81699,3,2865246930000104E7
7,85029,3,433948877999994E7
8,84709,3,4242546549999945E7
9,82005,3,3027247350000087E7
10,84848,3,4253045459999874E7
11,81731,3,286898035000004E7
12,84612,3,406825486000006E7
```

7.2.7 Identifying the Top-Selling Products

Groups products by name, sums the quantity sold, and shows the top 10 products by unit sales without letting high-priced products dominate.

7.2.7.1 Create the Pig file and Run the Pig script

```
C:\ECommerceBigData\pig>notepad pig7_topproducts.pig
C:\ECommerceBigData\pig>pig -x mapreduce pig7_topproducts.pig
2026-09-11 14:52:35,784 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 14:52:35,786 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 14:52:35,787 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 14:52:36,700 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 14:52:36,700 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789116756689.log
2026-09-11 14:52:37,736 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\USER/.pigbootstrap not found
2026-09-11 14:52:37,819 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 14:52:37,820 [main] INFO org.apache.pig.backend.hadoop.executionengine.MExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 14:52:38,798 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig7_topproducts.pig-7552e280-4ac2-4e96-8752-acf53d28b464
2026-09-11 14:52:38,799 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 14:52:40,202 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outputformat.separator
2026-09-11 14:52:40,235 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-11 14:52:40,270 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 14:52:40,315 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 14:52:40,493 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699480192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 14:52:40,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 14:52:40,658 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 14:52:40,694 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope: 217
2026-09-11 14:52:40,714 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-11 14:52:40,715 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-11 14:52:40,867 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 14:52:41,285 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 14:52:41,306 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 14:52:41,317 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 14:52:41,317 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.5
2026-09-11 14:52:41,322 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
```

7.2.7.2 Show the actual output

File contents	
Water Bottle Insulated,67853	
Vacuum Robot Cleaner,67703	
Tennis Balls Pack,67482	
Cotton T-Shirt Pack,67219	
Gym Gloves,67215	
Storage Box Set,67130	
LED Strip Lights 5m,67124	
Treadmill Folding,67109	

7.2.8 Identifying the Highest-Revenue Countries

Groups records by country, sums total_price_usd, sorts in descending order, and limits to the top 10 identifying the highest-revenue markets. This complements the per-country average, since a country can rank highly here simply by having a large number of orders even if its average order value is unremarkable.

7.2.8.1 Create the Pig file and Run the Pig script

7.2.8.2 Show the actual output

7.2.9 Brand Profitability Ranking

Groups brands by profit, sorts them in descending order, and identifies the most profitable brands regardless of sales volume.

7.2.9.1 Create the Pig file and Run the Pig script

7.2.9.2 Show the actual output

7.2.10 Delivery Performance Breakdown

Groups records by delivery status and calculates order count and average rating to compare delivery performance with customer satisfaction.

7.2.10.1 Create the Pig file and Run the Pig script

```
c:\ECommerceBigData\pig>notepad pig10_deliverystatus.pig
c:\ECommerceBigData\pig>pig -x mapreduce pig10_deliverystatus.pig
2026-09-11 15:16:31,866 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 15:16:31,869 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 15:16:31,871 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 15:16:32,728 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 15:16:32,728 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789118192716.log
2026-09-11 15:16:33,724 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file c:\Users\USER/.pigbootstrap not found
2026-09-11 15:16:33,827 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 15:16:33,827 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 15:16:34,606 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig10_deliverystatus.pig-a2f3176d-3e33-48d2-8d66-176d2dd5f14b
2026-09-11 15:16:34,607 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 15:16:35,774 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.t
extoutputformat.separator
2026-09-11 15:16:35,795 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 15:16:35,825 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 15:16:35,869 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED:[addForEach, columnMapKeyPrune, FilterConstantCalcul
ator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptim
izer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 15:16:35,974 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699408192 to monitor. collectionUsageThreshold
= 489580128, usageThreshold = 489580128
2026-09-11 15:16:36,065 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 15:16:36,085 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 15:16:36,122 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 15:16:36,123 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 15:16:36,263 [main] INFO org.apache.hadoop.yarn.client.RMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
```

7.2.10.2 Show the actual output

File contents

```
Failed,49994,2.9927191262951554
Pending,49774,3.0080765058062444
Delivered,750804,2.9993127367462082
In Transit,149551,3.00175191071942
```

8. Findings and Interpretation

MapReduce Results :

Job	Result
Job 1-Total Order Count	1000123
Job 2-Category-wise Sales	Clothing 4.5, Electronics 1.3 Health 5.7,Home 8.7,Sports 7.2
Job 3-Avg. Order Value by Country	~\$400–406 across all 10 countries
Job 4-Total Profit by Payment Method	~\$32.1M–\$32.4M across all 5 methods

Pig Results :

Analysis	Result
1. Row Count Check	Raw: 1,000,124 · Cleaned: 1,000,123
2. Total Order Count	1000123
3. Category-wise Sales	Samsung,8136060.9, Dell,8109250.9, Fitbit,8102206.1, Sony,8096025.2, GoPro,8083747, JBL,8074264.5, Philips,8067932.9, Xiaomi,8066665.4 Baseus,8064639, Huawei,8057126.3, Anker,8046326.7, Nike,8038814.4, Apple,8012777.9, Logitech,8012292.1, Nikon,8004462.2, Adidas,8002090.9, Canon,7998829, HP,7997523.5, Lenovo,7994543.6, Asus,7981233.5
4. Average Order Value by Country	Italy,405.7, Spain,403.6, Canada,401.9 France,401.7, Belgium,402.7, Germany,402.6, Australia,404.2, Netherlands,403.2, United States,403.4, United Kingdom,403.1
5. Payment Method Distribution	PayPal,199881,8.0, Apple Pay,199814,8.0, Debit Card,200805,8.1, Credit Card,199410,8.0, Bank Transfer,200213,8.0
6. Monthly Sales Trend	1,85100,3.4,2,77685,3.1,3,85079,3.4,4,82436,3.3,5,85100,3.4,6,81699,3.2,7,85029,3.4, 8,84799,3.4, 9,82005,3.3,10,84848,3.4,11,81731,3.2,12,84612,3.4
7. Top 10 Best-Selling Products	Water Bottle Insulated,67853, Vacuum Robot Cleaner,67703 Tennis Balls Pack,67482, Cotton T-Shirt Pack,67219 Gym Gloves,67215, Storage Box Set,67130 LED Strip Lights 5m,67124, Treadmill Folding,67109 Resistance Bands Set,67042, Desk Lamp LED,66928
8. Top 10 Countries by Revenue	Netherlands,4.0, Italy,4.0, Belgium,4.0, Spain,4.0, United Kingdom,4.03, Australia,4.03, United States,4.02, France,4.023, Canada,4.02, Germany,4.01
9. Brand-wise Total Profit	Clothing 4.5, Electronics 1.3, Health 5.7 Home 8.7, Sports 7.2
10. Delivery Status Breakdown	Clothing 4.5, Electronics 1.3 Health 5.7, Home 8.7, Sports 7.2

Comparing the Two Approaches

Both approaches produced consistent results for the analyses that overlapped — order count and country averages — confirming the reliability of the pipeline rather than any coincidence in one implementation or the other. However, Pig required substantially less code — each analysis was a handful of Pig Latin lines compared to three full Java classes (Mapper, Reducer, Driver) per MapReduce job — while still compiling down to

MapReduce jobs under the hood, meaning the productivity gain came without any real loss of processing power.

9. Limitations & Future Scope

Limitations

- Single-node Hadoop setup does not demonstrate true distributed performance gains.
- Batch processing cannot analyze new orders in real time.
- Basic cleaning only removes missing `order_id` and `total_price_usd` records.
- Raw text results make data interpretation harder for non-technical users.
- Manual execution makes rerunning the complete analysis time-consuming.

Future Scope

- Apache Hive can enable easier SQL-style querying for non-programmers.
- Apache Spark can provide faster in-memory processing for larger datasets.
- Multi-node Hadoop or AWS EMR can demonstrate real horizontal scalability.
- Power BI or Tableau can improve result visualization and interpretation.
- Oozie or Airflow can automate the complete data processing pipeline.

10. Conclusion

This project demonstrated the practical use of HDFS, MapReduce, and Apache Pig to store and analyse a large e-commerce dataset of over one million order records. The dataset was uploaded to HDFS and verified for integrity before any processing began, establishing a reliable foundation for everything that followed. Four MapReduce jobs, written as custom Java Mapper/Reducer/Driver classes, computed the total order count, category-wise sales, average order value by country, and total profit by payment method. Ten Apache Pig scripts then extended this analysis to payment distribution, monthly trends, best-selling products, top-revenue countries, brand profitability, and delivery performance, using Pig Latin to express the same style of aggregation far more concisely than the equivalent MapReduce code would have required.

The results show that average order value and profit are both fairly evenly distributed across countries and payment methods respectively, and that the two Pig/MapReduce implementations of the same analysis, average order value by country, produced consistent figures, which validates the overall pipeline rather than any single job in isolation. This project reinforces how distributed Big Data tools can turn a large, real-world transactional dataset into structured business insight, and lays a foundation for future extensions using Hive or Spark, or for scaling the same approach to a genuinely distributed, multi-node cluster.

11. Reference

<https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop>